Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2007-I-Q06] | Part B (Q6-Q10)

Answer: 169

1. Split by whether the sequence visits 13 and 26. Between successive landmarks $0, 13, 26, 39$, there are five routes: climb successive multiples of three, then jump to the next landmark at one of five possible starting positions.

2. If the intermediate landmark is omitted, each corresponding skip segment has four routes. Thus paths visiting both landmarks number $5^3 = 125$, only either one contribute $5 \cdot 4 = 20$ each, and neither contributes four.

3. Total $125 + 20 + 20 + 4 = 169$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2007-II-Q06] | Part B (Q6-Q10)

Answer: 640

1. Build from right to left. A final digit d may be preceded by an odd digit below d or an even digit above d. For every $d = 0, \ldots, 9$, these possibilities total exactly four; zero is never an allowed predecessor.

2. The last digit has ten choices and each preceding digit four, with a nonzero leading digit automatic. Count $10 \cdot 4^3 = 640$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2013-II-Q09] | Part B (Q6-Q10)

Answer: 106

1. With k available colors, color the first tile in k ways. At each of six following cells, either continue the current tile or start a new tile in any of k colors. This gives $k(k+1)^6$.

2. Inclusion-exclusion for missing colors yields $N = 3 \cdot 4^6 - 3(2 \cdot 3^6) + 3(2^6) = 8106$.

3. The required remainder is 106.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2015-II-Q10] | Part B (Q6-Q10)

Answer: 486

1. Let T_n be the count. Insert the maximum n into an admissible permutation of $1, \ldots, n-1$. The entry immediately after it, if present, must be $n-1$ or $n-2$. Hence there are exactly three insertion positions: before those entries or at the end.

2. Every insertion preserves all other inequalities. Conversely, deleting n is safe: its following entry is at least $n-2$, while its preceding entry is at most $n-1$, so their newly formed adjacent inequality still holds.

3. Thus $T_n = 3T_{n-1}$ for $n \geq 3$, with $T_2 = 2$. Therefore $T_7 = 2 \cdot 3^5 = 486$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-II-Q08] | Part B (Q6-Q10)

Answer: 556

1. For either coordinate, a multiset of jumps totaling four is of type $(1,1,1,1)$, $(1,1,2)$, or $(2,2)$. Choose one type independently for horizontal and vertical jumps, and interleave their distinct directional steps.

2. The symmetric table of multinomial counts for these three types is $\begin{pmatrix} 70 & 105 & 15 \\ 105 & 180 & 30 \\ 15 & 30 & 6 \end{pmatrix}$. For example the middle entry is $6!/(2!2!) = 180$, since the two horizontal singles and two vertical singles are indistinguishable within each direction.

3. Summing all entries gives 556.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2024-I-Q06] | Part B (Q6-Q10)

Answer: 294

1. Four turns mean five nonempty runs. A path beginning right has run directions R, U, R, U, R.

2. The three horizontal run lengths are positive integers summing to 8, giving $\binom{7}{2}$ choices; the two vertical lengths give $\binom{7}{1}$.

3. Interchanging horizontal and vertical accounts for paths starting up. Hence $2\binom{7}{2}\binom{7}{1} = 294$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2000-I-Q15] | Part C (Q11-Q15)

Answer: 927

1. Assign the original positions $1, \ldots, 2000$ to a queue. Repeatedly remove the front position and rotate the next position to the back. The removed positions receive labels in increasing order.

2. For a reproducible recurrence, let $L = [1, \ldots, 2000]$ and repeat $p = L.pop(0)$, followed, when nonempty, by $L.append(L.pop(0))$. Recording the last two removals gives $928, 1952$. This finite procedure tracks positions exactly and uses no assumptions about card labels.

3. Thus label 1999 began at position 928, with 927 cards above it.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2001-I-Q14] | Part C (Q11-Q15)

Answer: 351

1. Encode receipt by 1 and nonreceipt by 0. Track counts A_n, B_n, C_n of strings ending respectively in 1, exactly one trailing zero, or exactly two trailing zeros.

2. The updates are $(A_{n+1}, B_{n+1}, C_{n+1}) = (B_n + C_n, A_n, B_n)$, starting from $(1, 1, 0)$ at length one.

3. The total counts for lengths 13 through 19 are $65, 86, 114, 151, 200, 265, 351$. Thus there are 351 patterns.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2004-I-Q15] | Part C (Q11-Q15)

Answer: 511

1. Build exact first-hit levels backwards. Start $T_1 = \{1\}$. Each z has predecessor $10z$, and also $z-1$ precisely when $z > 2$ and $z \not\equiv 1 \pmod{10}$. The latter exclusions prevent an incorrect divide-by-ten predecessor and a return from the absorbing state one.

2. These predecessor sets are disjoint because f has a unique output. Iterating this explicit rule gives level sizes at $n = 10, \ldots, 20$:
$256, 511, 1022, 2043, 4084, 8164, 16320, 32624, 65216, 130368, 260608$.

3. Thus $M = 260608 = 2^9 \cdot 509$. Trial division through $\sqrt{509} < 23$ proves 509 prime, and the required sum is 511.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2005-I-Q13] | Part C (Q11-Q15)

Answer: 083

1. Track counts by the last step, $R(x,y), U(x,y), D(x,y)$. Then $R(x,y) = R(x-1,y) + D(x-1,y)$, $U(x,y) = U(x,y-1) + D(x,y-1)$, and $D(x,y)$ is the sum of all three states at $(x-1,y-1)$.

2. Use one empty starting state and allow all three first steps; along each axis the only path uses that axis direction. Filling the recurrence gives $(D,R,U)(5,5) = (27,28,28)$.

3. The desired total is $27+28+28 = 83$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2006-I-Q11] | Part C (Q11-Q15)

Answer: 458

1. In a legal tower using $1, \ldots, n$, the new largest cube $n+1$ can be inserted only at bottom, directly above n, or directly above $n-1$: exactly three choices for $n \geq 2$. Each insertion stays legal.

2. Conversely deleting $n+1$ preserves legality because its supporting cube had side at least $n-1$, and the cube above it has side at most n. Thus $T_{n+1} = 3T_n$, with $T_2 = 2$.

3. Therefore $T_8 = 2 \cdot 3^6 = 1458$, remainder 458.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2008-I-Q11] | Part C (Q11-Q15)

Answer: 172

1. Let a_n, b_n count valid words starting with A, B. Removing the initial AA gives $a_n = a_{n-2} + b_{n-2}$ for $n \geq 3$. For a B start, either remove a single initial B before an A run, or remove BB from a longer odd B run; hence $b_n = a_{n-1} + b_{n-2}$.

2. Initialize $(a_1, b_1) = (0, 1)$ and $(a_2, b_2) = (1, 0)$. Iteration gives $(a_{10}, b_{10}) = (16, 21)$, $(a_{12}, b_{12}) = (37, 43)$, and $(a_{14}, b_{14}) = (80, 92)$.

3. The desired total is 172.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2015-II-Q12] | Part C (Q11-Q15)

Answer: 548

1. Fix the first letter. A valid string is uniquely specified by its successive run lengths, each in $\{1, 2, 3\}$ and summing to 10. The letters alternate automatically.

2. Let C_n count these compositions, with $C_0 = 1$ and $C_n = 0$ for $n < 0$. Removing the last part gives $C_n = C_{n-1} + C_{n-2} + C_{n-3}$.

3. The values for $n = 1, \ldots, 10$ are $1, 2, 4, 7, 13, 24, 44, 81, 149, 274$. Either first letter is possible, so the answer is $2 \cdot 274 = 548$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2016-II-Q12] | Part C (Q11-Q15)

Answer: 732

1. Let R_n count proper colorings of an n-cycle. A proper row of n positions has $4 \cdot 3^{n-1}$ colorings. Rows with different end colors correspond to R_n; those with equal ends merge to a properly colored $(n-1)$-cycle.

2. Thus $R_n + R_{n-1} = 4 \cdot 3^{n-1}$. From $R_3 = 24$, obtain $R_4 = 84$, $R_5 = 240$, $R_6 = 732$. The six-region answer is 732.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-I-Q14] | Part C (Q11-Q15)

Answer: 351

1. Group nonabsorbing vertices into $A = \{S, P_1\}, B = \{P_2, P_5\}, C = \{P_3, P_4\}$. Let (a_n, b_n, c_n) count still-active paths in these classes after n steps.

2. The transitions give $(a_{n+1}, b_{n+1}, c_{n+1}) = (a_n + b_n, a_n + c_n, b_n)$, initially $(1, 0, 0)$. Exactly c_n paths first reach E next.

3. For $n = 0, \ldots, 11$, the c_n values are $0, 0, 1, 1, 3, 4, 9, 14, 28, 47, 89, 155$. Their sum is 351.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-II-Q11] | Part C (Q11-Q15)

Answer: 461

1. Failure means a proper prefix contains exactly the first k positive integers. Let c_n count permutations of $1, \ldots, n$ with no such proper prefix.

2. Every permutation has a unique earliest closed prefix, of some size k. Its internal order has c_k choices, and the rest are arbitrary, so $n! = \sum_{k=1}^{n} c_k (n-k)!$.

3. Starting with $c_1 = 1$ yields $c_2 = 1, c_3 = 3, c_4 = 13, c_5 = 71$. Then $c_6 = 720 - (120 + 24 + 18 + 26 + 71) = 461$.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2019-II-Q12] | Part C (Q11-Q15)

Answer: 047

1. Let $g(s)$ count such sequences of sum s with first term at least two. Set $g(1)=0$. For $s \geq 2$, the single-term sequence contributes one. If first term is a proper divisor $d \geq 2$ of s, divide the remaining sequence by d; it has sum $s/d-1$ and first term at least two. Thus $g(s)=1+\sum_{d \mid s,\, 2 \leq d<s} g(s/d-1)$.

2. The desired count is $1+\sum_{d \mid 360,\, d<360} g(360/d-1)$. There are twenty-three arguments. Fifteen have value one, one argument is one and contributes zero, while arguments $119, 44, 39, 35, 14, 9, 8$ have values $6, 8, 6, 4, 3, 2, 2$.

3. The sum is $1+15+6+8+6+4+3+2+2=47$. The recursion only calls smaller positive sums, so it gives a finite exhaustive count.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2023-I-Q11] | Part C (Q11-Q15)

Answer: 235

1. Let T_m count subsets of a block of m consecutive integers with no adjacent pair. Choosing or omitting its last element gives $T_m = T_{m-1} + T_{m-2}$, with $T_0 = 1, T_1 = 2$. Thus $T_m = F_{m+2}$.

2. If the unique pair is $k, k+1$, forbid $k-1, k+2$. The blocks on either side contribute $F_k F_{10-k}$; at an endpoint, the empty block contributes 1.

3. For $k = 1, \ldots, 9$, these contributions are $34, 21, 26, 24, 25, 24, 26, 21, 34$. Their sum is 235.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2026-I-Q15] | Part C (Q11-Q15)

Answer: 083

1. Enumerate every rectangle $[r, r+a) \times [c, c+b)$, keeping its boundary-cell bitmask. There are $(10 \cdot 9/2)^2 = 2025$ candidate loops. A partition is an exact cover of the 100 cells by five masks.

2. Use the following exact recurrence: $C(U, j) = \sum C(U \cup L, j-1)$, where v is the first uncovered cell and the sum is over candidate loops L containing v with $L \cap U = \varnothing$. Set $C(U, 0) = 1$ exactly for the full grid, otherwise zero. This counts each partition once, because its loop covering v is unique.

3. Evaluating the recurrence gives $C(\varnothing, 5) = 83$. The count splits into 27 partitions with a full 10×10 outer loop, 28 with multiple vertical outer slabs, and 28 with multiple horizontal outer slabs. The companion exact-cover certificate records these masks and totals.

Source / 来源

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2026-II-Q14] | Part C (Q11-Q15)

Answer: 157

1. The running-value parity equals the ordinary prefix-sum parity, since adding and subtracting agree modulo two. Hence the final value is 12 minus twice the sum of even terms placed after an odd prefix; those subtracted terms must sum to six.

2. For a completely explicit count, define $D(s, v)$ as the number of positive-part sequences of total s with running value v, starting at $D(0, 0) = 1$. For every $1 \le t \le 12 - s$, add $D(s, v)$ to $D(s + t, v - t)$ if v is odd and t even, and otherwise to $D(s + t, v + t)$. This is an exhaustive disjoint extension recurrence.

3. Evaluating by total gives $D(12, 0) = 157$. Equivalently, grouping by the number $2m$ of odd terms and their total A, the six nonzero contributions $(m, A) = (1, 2), (1, 4), (1, 6), (2, 4), (2, 6), (3, 6)$ are $20, 16, 12, 36, 48, 25$, summing to 157.

Source / 来源